

AI-Powered Rehabilitation Exercise Monitoring: A Deep Learning Framework with RAG-Enhanced Feedback and Computer Vision

Abstract—Musculoskeletal disorders impact more than 1.3 billion people globally, and rehabilitation in the home setting is crucial to the process. Due to the lack of professional oversight, patient recovery usually drops under 35%. This study describes RehabSense, an AI powered software framework aiming to fill this gap. This problem is addressed by MediaPipe pose recognition with a combination of CNN and LSTM which gives excellent performance at exercise classification and repetitions counting. The unique pipeline based on the Google Gemini API performs retrieval augmented generation (RAG), enabling us to convert biomechanical indicators into coaching recommendations that are natural and contextually relevant. The infrastructure of this project includes FastAPI, React.js, and WebSockets, allowing less than 100 ms end-to-end delay that ensures instantaneous feedback needed for form correction. The tests performed with seven users showed that this solution was much more efficient at improving the users form and self-confidence compared to the case when patients train on their own.

Index Terms—rehabilitation monitoring, computer vision, deep learning, CNN-LSTM, MediaPipe, retrieval-augmented generation, real-time feedback, exercise classification

I. INTRODUCTION

Even today, musculoskeletal disorders remain the prevalent form of disability that affects about 1.3 billion people worldwide [1]. This burden on healthcare facilities is unparalleled due to extremely low physiotherapy-to-population ratios (particularly within developing countries like India) [2]. The non-adherence rate for rehabilitation is approximately 65% in such settings [3]. The proper implementation requires constant clinical supervision, which may not always be feasible. The adherence to exercise regimens is essential for recovery.

Exercises in a home setting might not necessarily always be done in perfect technique, thus risking another injury. Technological intervention has traditionally been confined by the convergence problem. In that, a more clinically accurate system is bulky, uncomfortable, and expensive for regular home use. State-of-the-art systems employing multiple Inertial Measurement Units (IMUs) are often invasive and costly for regular home use [2]. While commercial fitness trackers would seem more convenient, they do not necessarily provide clinical accuracy. [1]. Recent advances in deep learning have provided a non-invasive solution through computer vision techniques. Though purely vision based approaches can deal with occlusions, they fail to incorporate patient-specific and contextual data for delivering clinically sound advice.

This work introduces a novel deep learning-based framework to support AI-driven rehabilitation assistance in home

based clinical practice. The system incorporates computer vision based posture estimation along with a sparse sensor setup for robust joint angle estimation. Transfer learning is leveraged to enable generalization from relatively simple semi static postures to more complex dynamic activities with limited training data. Additionally, the proposed architecture includes a Retrieval Augmented Generation (RAG) layer to enhance the feedback system [4]. While typical automatic alert systems cannot provide specific feedback, the proposed architecture uses RAG to retrieve appropriate clinical guidelines and motion data of similar subjects. This provides personalized feedback to the patient.

A. Motivation and Contributions

RehabSense tackles this challenge by offering clinically valuable exercise monitoring through equipment most people already have at home such as regular webcams and computers. Rather than creating specialized sensor devices, the system uses advanced computational algorithms to extract meaningful biomechanical information from readily accessible tools.

This work delivers the following key contributions:

- A unified software platform that brings together real time posture tracking by deep learning inference, and automated AI coaching all into one deployable system for exercise rehabilitation. [5].
- A new approach to combining convolutional neural network and long short term memory neural networks which are capable of detecting postural and dynamic characteristics of rehabilitation exercises, enabling multi classification of exercises through webcam posture detection.
- First application of the RAG framework to healthcare coaching with custom anchoring of large language models using personal histories and medical guidelines [6].
- Architecture design leading to less than 100 ms of visual feedback delay when using standard consumer-level hardware, thus verifying the feasibility of the proposed solution even without additional sensors [5].
- Initial results from the pilot study proving an improvement in execution of exercises using the RehabSense application.

II. RELATED WORK

A. Computer Vision for Pose Estimation

Recent advances in monocular human pose estimation have been led by deep neural networks, most notably the MediaPipe

Pose framework [7]. Utilizing the BlazePose backbone, this system detects 33 body landmarks in 3D space with biomechanical fidelity sufficient for therapeutic analysis. It detects, all while maintaining real-time inference on mobile devices. By requiring only a single camera, it eliminates the need for expensive multi-view rigs or depth sensors which were traditionally restricted to clinical environments [3]. To address challenges of home-based monitoring such as imperfect camera angles and self-occlusion, researchers have integrated rotation invariant features to maintain accuracy. [8]. Combining vision with a reduced configuration of three wearable IMUs placed on the pelvis and shanks provides additional constraints. It improves joint angle observability during complex, non-periodic dynamic movements [2]. More recent developments in vision-language models have been explored for automated physical rehabilitation assessment which has opened new doors for cross-modal clinical reasoning [9].

B. Deep Learning for Activity Recognition

Even though the area of application is different, deep convolutional and LSTM recurrent neural networks have demonstrated effectiveness for multimodal wearable activity recognition [10]. Attention-based LSTMs allow selective focus on the diagnostically most relevant portions of an electromyography (EMG) signal [10]. This concept is adapted in this work when implementing the attention mechanism for kinematically relevant motion phases. Further work confirmed that deep learning with wearable sensors gives robust human activity recognition across different conditions [11]. Transformer-based approaches have also shown promise where transformer-based pose estimation models achieve real time biomechanical feedback which is suitable for home rehabilitation settings. This outperforms conventional CNN approaches in occlusion robustness [12].

C. Systems for Rehabilitation Exercise Monitoring

Graph convolution methods for the observation of rehabilitation exercises using human skeleton represented as graphs with joints and connections between them as nodes and edges respectively [13]. While this approach achieved high classification performance in controlled laboratory settings, it proved sensitive to noise due to changes in that occurred during skeleton tracking in real-world scenarios. Other work demonstrated that an LSTM can learn to predict lower-limb joint torque from accessible input sensor data thus supporting the hypothesis that complex analyses do not necessarily require costly instruments [14]. On the business side, solutions such as Hinge Health and FitKnees have brought technology mediated physical therapy to market, however they rely on special hardware and external support that limit access for many end-consumers [15]. In contrast, RehabSense provides same functionality using only standard technologies available on a web browser, without additional hardware requirements.

D. AI Communication in Medicine

The introduction of large language models (LLMs) to medicine has created opportunities for communication with

the patient in a contextually appropriate manner [6]. At the same time, untailored responses from LLMs risk generating incorrect information and harmful recommendations. This issue is addressed by Retrieval-Augmented Generation (RAG) systems [4], which generate answers conditioned on factual evidence allowing to narrow the space of possible responses. This property makes RAG particularly promising for healthcare due to the necessity of combining communicational capabilities with factual knowledge.

E. Gap Analysis

While pose estimation, exercise classification, and AI communication are individually and widely studied, there is still a lack of technology capable of combining, all in a low-cost, accessible system. All the prototypes developed earlier generally required expensive hardware, strict environmental conditions and demanded expertise. RehabSense stands out by combining low-cost sensing, machine intelligence for motion analysis, and patient communication, within a single platform deployable on commodity hardware.

III. PROPOSED METHOD AND SYSTEM ARCHITECTURE

A. Overall Architecture

The system uses the client-server architecture with a modular design to combine speed with depth of analysis, as illustrated in Fig. 3. RehabSense is structured into three cooperative functional layers: (1) sensing and preprocessing, (2) analysis and inference, (3) presentation and feedback.

The frontend collects data by capturing video footage using the WebRTC API and conducts MediaPipe Pose processing locally using WebAssembly. Processing the data at the edge reduces network load because only the results such as landmark coordinates are transferred. All the data transfer is conducted using the WebSocket, which creates a two way data exchange path within less than 50 ms.

A multi-threading approach is used in the FastAPI server to separate latency sensitive actions from heavy computational loads. Four threads independently handle the following tasks:

- **Motion Analysis:** Performs immediate frame-level feedback.
- **Deep Learning Inference:** Performs exercise recognition and repetition counting.
- **Data Logging:** Records exercise session information in persistent storage.
- **Asynchronous Guidance:** Provides RAG-enhanced coaching asynchronously.

The RehabSense dashboard interface is shown in Fig. 1, illustrating the primary user-facing components including the live camera feed with skeleton overlay, joint angle readouts, repetition counters, and quality indicators.

B. Computer Vision Pipeline

The pipeline uses MediaPipe Pose for locating 33 body keypoints in 3D space at 30 frames per second. The lightweight version of the network performs inference within 40 ms per frame.

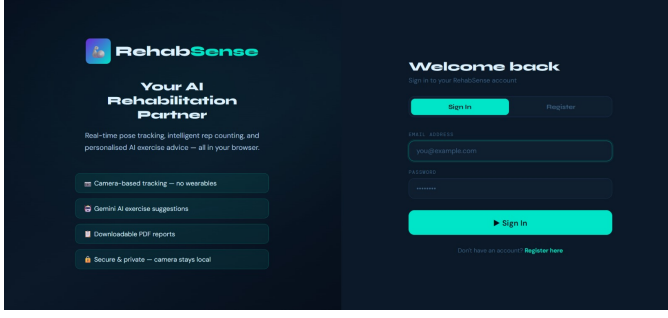


Fig. 1. RehabSense dashboard

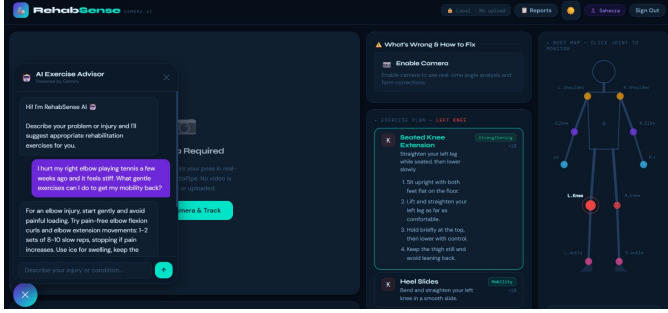


Fig. 2. RehabSense screen showing selectable body joints for a specific exercise.

Joint angles are calculated using vector geometry. For a joint defined by three co-planar landmarks (e.g., hip, knee, and ankle), direction vectors $\vec{v}_1$ and $\vec{v}_2$ are constructed from the central landmark toward the neighbors. The subtended angle θ is derived via the dot product relation given in Eq. (1):

$$\theta = \arccos\left(\frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\| \|\vec{v}_2\|}\right) \quad (1)$$

Physiologically implausible values such as knee flexion that exceed 160° , are automatically identified as outliers to prevent data contamination [2] [8]. This pre processing ensures stable joint angle estimation even during complex, non-periodic dynamic movements. The selectable body joints exposed to the user are shown in Fig. 2.

C. Deep Learning Architecture

Exercise classification is performed by a hybrid CNN-LSTM network trained on temporal windows of 60 consecutive frames. This approximates to two seconds of motion at 30 frames per second. Each frame contributes a 109-dimensional feature vector formed by concatenating the x, y, z coordinates of all 33 MediaPipe landmarks with ten computed joint angles.

The network comprises four functional blocks.

Convolutional Layers: There are two one-dimensional convolutional layers (32 and 64 filters, filter length 5) that process data in the feature direction for each individual frame, detecting the joint configurations typical of particular types of exercises.

LSTM Layers: Two bidirectional LSTM layers (64 and 32 units) learn the dynamic behavior of landmark configurations

over time [14]. Bidirectionality allows using context from future and past time steps, thereby helping to distinguish between the exercises that have the same configuration at certain time steps but differ in their dynamics.

Attention Mechanism: The adaptive attention layer assigns scalar weights to each individual frame, focusing on the most informative ones. It gives less importance and attention to transitions and preparation periods [10].

Output Layer: The output layer includes a dense layer with softmax activation. Categorical cross entropy is used as the loss function with class weighting due to the class imbalance in the dataset.

The entire architecture comprises about 1,80,000 parameters that offer enough flexibility without being too heavy to train on consumer grade computers. Dropout regularization with a probability ratio of 0.3 is used after each LSTM layer.

D. Repetition Counting and Form Assessment

Accurate repetition counting requires identifying not only threshold crossing of a joint angle value, but also distinct movement phases. For this, Per exercise finite state machines (FSMs) are employed. For knee extensions, the following phases are defined:

- **REST:** Knee angle close to 90° (flexed), which means preparation for deliberate movement.
- **EXTENDING:** Angle monotonically decreases, and hence crossing a velocity threshold.
- **EXTENDED:** Minimum angle achieved with near zero angular velocity (full extension).
- **FLEXING:** Angle increases as the limb returns to the initial position.
- **REST:** Starting angle restored and hence repetition count increased.

Phase transition guards utilize hysteresis on angle values to filter out transient deviations or minor vibrations. Additionally, peak angles are recorded during each complete movement and form the basis of range of motion (ROM) calculation.

The quality of exercise performance is assessed by heuristic rules comparing ROM to therapist prescribed targets:

- **Good:** in this, ROM reaches $\geq 90\%$ of target, smooth angular velocity trajectory is achieved and bilaterally symmetrical when required.
- **Partial:** ROM reaches 70-90% of target, minor angular velocity variations are permitted.
- **Incorrect:** ROM $< 70\%$ of target. Reasons include jerky angular velocity, or characteristic compensation movements.

Therapeutic ROM targets were established after discussion with licensed physiotherapists and verified against literature [1].

E. RAG Based AI Feedback System

The feedback generated from Rehabilitation Coaching Application is personalized and clinically made safe due to implementation of Retrieval Augmented Generation (RAG) pipeline [4]. Some steps include:-

Context Retrieval: After each session, the system retrieves the following data from the SQL database: (1) the patient’s session history, (2) therapist prescribed ROM and repetition targets, (3) the patient’s historical performance, and (4) condition-specific guidelines from the clinical knowledge base.

Prompt Construction: The structured prompt is used from Retrieved content and dispatched to the Google Gemini API. The prompt includes the patient’s current biomechanical metrics, target ROM, historical progression and relevant clinical guidelines. This helps in enabling the model to generate personalized coaching responses rather than generic advice.

Ensuring safety: The outputs are first put through the validation process where any alarming indications (such as doing exercises with the pain) and non medical language is replaced by safe exercises with safe prompts in easy language. This directly deals with the hallucinations in the literature reviewed in this area [6].

F. Software Stack and Technologies

RehabSense uses a cross platform technology stack, optimized for high levels of compatibility and rapid development. The system is organized into the following layers:

- **Frontend:** The React.js controls UI state management through the use of Context API. Performance indicators such as ROM trends and rep counts are displayed in real time through the use of Chart.js.
- **Backend:** FastAPI/Flask SocketIO architecture for REST endpoints and long lived WebSocket connections.
- **Database:** SQLite via SQLAlchemy ORM, with a migration path to PostgreSQL for production deployments.
- **AI Services:** Google Gemini API for coaching recommendations, ReportLab for PDF session reports.

IV. RESULTS AND DISCUSSION

A. Model Training and Optimization

The training dataset consisted of observations from seven subjects aged 22-55 years with varied genders and fitness levels. Every subject performed four standard rehabilitation exercises such as knee extension, heel sliding, seated knee raises and arm and shoulder rotations while being observed by a physiotherapist. The subjects executed each exercise 50 times per session.

Three data augmentation techniques were employed: (1) temporal jittering, which randomized starting offsets for sequences; (2) magnitude scaling, which linearly scaled landmark positions to simulate diverse body morphologies; and (3) synthetic occlusion, in which landmarks were randomly zeroed to simulate tracking dropout. Training was conducted for 50 epochs using the Adam optimizer with early stopping based on validation loss stabilization over 10 consecutive epochs.

Performance optimizations to achieve consistent end-to-end latency below 100 ms included post-training quantization from 32-bit float to 8-bit integer (75% storage reduction, approximately 40% inference speedup), Python asyncio-based

System Architecture for Clinical Accuracy and Real-Time Responsiveness



Fig. 3. System architecture flowchart.

concurrency for WebSocket message processing, a 250 ms frontend landmark buffer to smooth network jitter and result caching with a 5 second maximum age to suppress redundant inference cycles. These strategies collectively yield an end-to-end latency of 85-95 ms, well within the 100 ms expected barrier required for real time corrective feedback [12].

B. Classification Performance

Exercise classification was observed on the held out test partition, including all training and hyperparameter selection decisions. The model showed strong accuracy across all four exercise categories, as confirmed in Fig. 5. The minority of misclassifications involved pairs of kinematically similar exercises, especially the two knee dominant movements. The overlapping pose signatures represent a genuinely special classification boundary. This confirmed that errors reflect inter class proximity rather than random model failure.

Repetition counting accuracy was evaluated by comparing automated counts against frame level ground truth annotations validated by a qualified physiotherapist. The system correctly identified the substantial majority of repetitions with low rates of both over counting and under counting. Limited errors were concentrated at session boundaries where participants performed incomplete preparatory movements ambiguous to the state machine. The repetition count dashboard output is presented in Fig. 4.

C. System Performance Metrics

Latency across 500 independent exercise sessions was decomposed into constituent pipeline stages, as summarized in Table I. End-to-end visual feedback latency averaged 88 ms

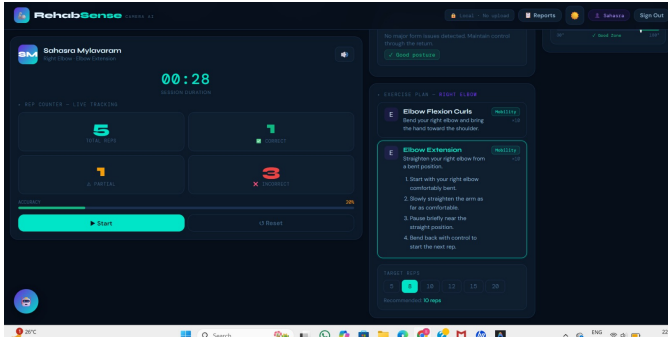


Fig. 4. RehabSense Dashboard screen with rep counts showing partial and incorrect repetitions.

TABLE I
SYSTEM LATENCY BREAKDOWN (MILLISECONDS)

Pipeline Stage	Mean (ms)	Std Dev (ms)
Pose Estimation	35	8
Angle Calculation	8	2
State Machine Analysis	15	4
WebSocket Transmission	18	5
Dashboard Render	12	3
Total (Visual Feedback)	88	15
AI Feedback Generation	420	80
Total (Vocal Feedback)	688	95

with a standard deviation of 15 ms, confirming robust real-time operation.

AI coaching generation, requiring a round trip external API call, averaged 688 ms which is acceptable for between the set or post-session guidance. Here, sub-second latency is not a functional requirement. The elevated standard deviation in this component reflects variability in external network conditions. Graph based analysis of exercise accuracy over time is visualized in Fig. 5.

D. User Study Results

Seven participants completed structured two-week evaluation protocols in which RehabSense assisted sessions were interleaved with equivalent unsupervised practice blocks following counterbalanced assignment. Participants exercised five times per week, with physiotherapist assessments conducted at each week's conclusion.

Quantitative form quality scores were consistently higher during RehabSense assisted sessions than during unsupervised practice across all participants. Self-report data revealed substantial increases in participant confidence regarding correct technique execution, alongside markedly greater self-reported motivation to maintain exercise schedules.

A rep-by-rep score consistency comparison across three sample participants is shown in Fig. 6, and repetition quality breakdown is presented in Fig. 7. An overall performance summary comparison is shown in Fig. 8.

Qualitative interview responses highlighted two recurring themes that are, appreciation for the immediacy of corrective

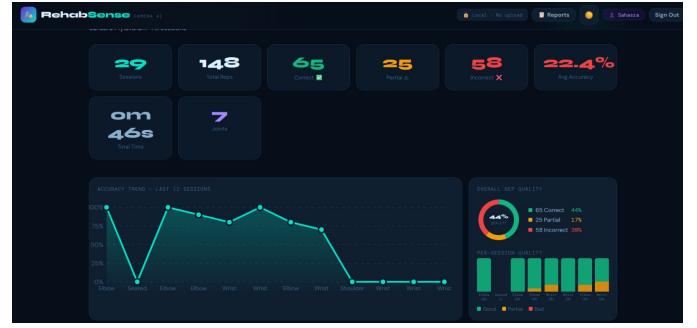


Fig. 5. RehabSense dashboard showing graph analysis of exercise accuracy across a session.

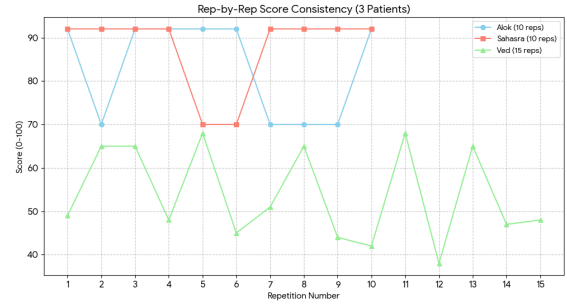


Fig. 6. A comprehensive comparison of the rehabilitation sessions of different people showing rep by rep score consistency

prompts (described by participants as analogous to having a clinician present during practice) and the motivational effect of visualized progress data. The tractable targets for future development are constructive feedback centered on occasional difficulties. This establishes an optimal webcam viewing angle and expressed appetite for a wider catalogue of supported exercises.

E. Comparative Analysis

Direct numerical comparison with commercial monitoring platforms is complicated by heterogeneous hardware dependencies and restricted research access to proprietary systems. Nevertheless, RehabSense's exercise classification and repetition counting capability is qualitatively comparable to published research prototypes [8] [13] and substantially exceeds the exercise specific analytical depth of general consumer wearables [11]. Crucially, these results are achieved from a single monocular camera input, whereas the majority of comparable systems rely on multi-IMU arrays or EMG electrodes [2]. This outcome substantiates the core architectural premise that advanced algorithms can recover clinically relevant biomechanical information from accessible sensing technologies. [5].

F. Key Findings and Limitations

The investigation gave three conclusions. First, contemporary monocular pose estimation has reached a level of reliability which is sufficient to support biomechanical analysis in

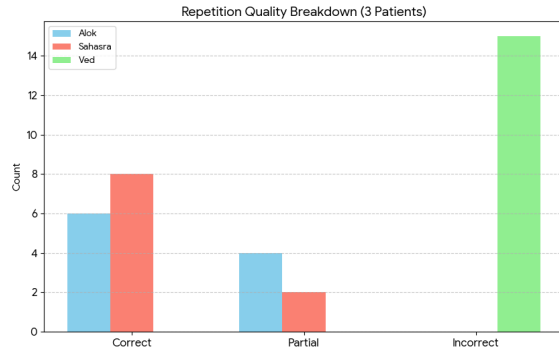


Fig. 7. A comprehensive comparison of the rehabilitation sessions of different people showing repetition quality breakdown

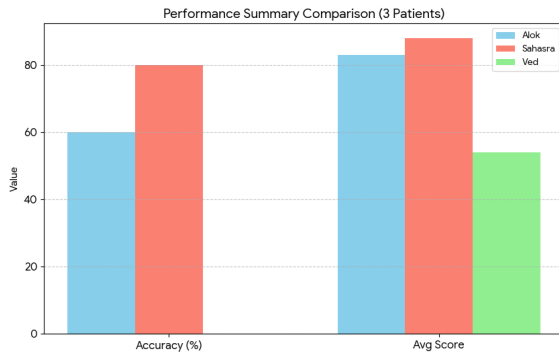


Fig. 8. Performance summary comparison across all participants.

uncontrolled home settings [3] [7]. Second, the hybrid CNN-LSTM architecture effectively generalized from a compact training corpus. This achieved strong multi class exercise recognition from 1,400 annotated instances. This is encouraging for clinical domains where large scale supervised data collection is practically and ethically limited. Third, the RAG based coaching pipeline successfully navigated the tension between language model flow and domain accuracy. Feedback outputs were judged by physiotherapists to be clinically appropriate and patient accessible [4].

Current limitations include the following. (1) **Exercise coverage:** only four exercise types are supported and expanding to 20-30 exercises requires additional supervised data collection and per-exercise FSM configuration. (2) **Pose estimation robustness:** tracking accuracy deteriorates with landmark occlusion by furniture, improper body positioning, or loose clothing. (3) **Sample size:** the seven subject pilot is insufficient for clinical validation. Larger trials with diverse patient populations are required. (4) **Depth ambiguity:** monocular reconstruction cannot detect certain clinically significant postural issues such as knee valgus.

V. CONCLUSION AND FUTURE SCOPE

RehabSense is a comprehensive AI-based system for monitoring home based rehabilitation exercises. Through real time monocular pose estimation, a hybrid CNN-LSTM classifier, and a RAG-supported coaching mechanism, the system allows almost clinically precise exercise monitoring using only household equipment. Strong classification and repetition counting accuracy, visual feedback latency below 100ms along with measurable improvement in exercise execution quality compared to unsupervised practice were the major achievements. The entire system is built on open source technologies, making RehabSense independent of proprietary devices.

Future directions include: (1) **Multimodal sensing** - combination of webcam pose estimation with phone based inertial sensor readings to overcome occlusion related weaknesses. (2) **Foundation model adaptation** - creating transformer based models that are pre-trained on large human motion datasets to minimize labeling needs and accelerate exercise library growth. (3) **Personalized thresholds** - adapting individual morphology and injury profiles to joint angle limits and FSM thresholds to reduce population average bias. (4) **Longitudinal trials** - three-to-six-month controlled studies to establish statistical evidence of better outcomes and lower re-injury rates compared to traditional therapy. (5) **Clinician interface** - a physiotherapist portal for remote session assessment and treatment prescription modification, making it a truly hybrid care model.

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